ABC Phones

Credit Portfolio:
From Raw Snapshots to Reliable Insights

133%

Loan-book growth across 2025 (8.8K → 20.5K loans) but credit quality deteriorating

44.8%

Latest delinquency rate — collection rate fell 10pp from Q1 to Q4 (69 → 60%)

+5pp

Default-rate gap for loans without complete customer profiles vs the rest of the book

What we found in the data

Three sources, real-world inconsistencies, ~50% customer-side coverage gap

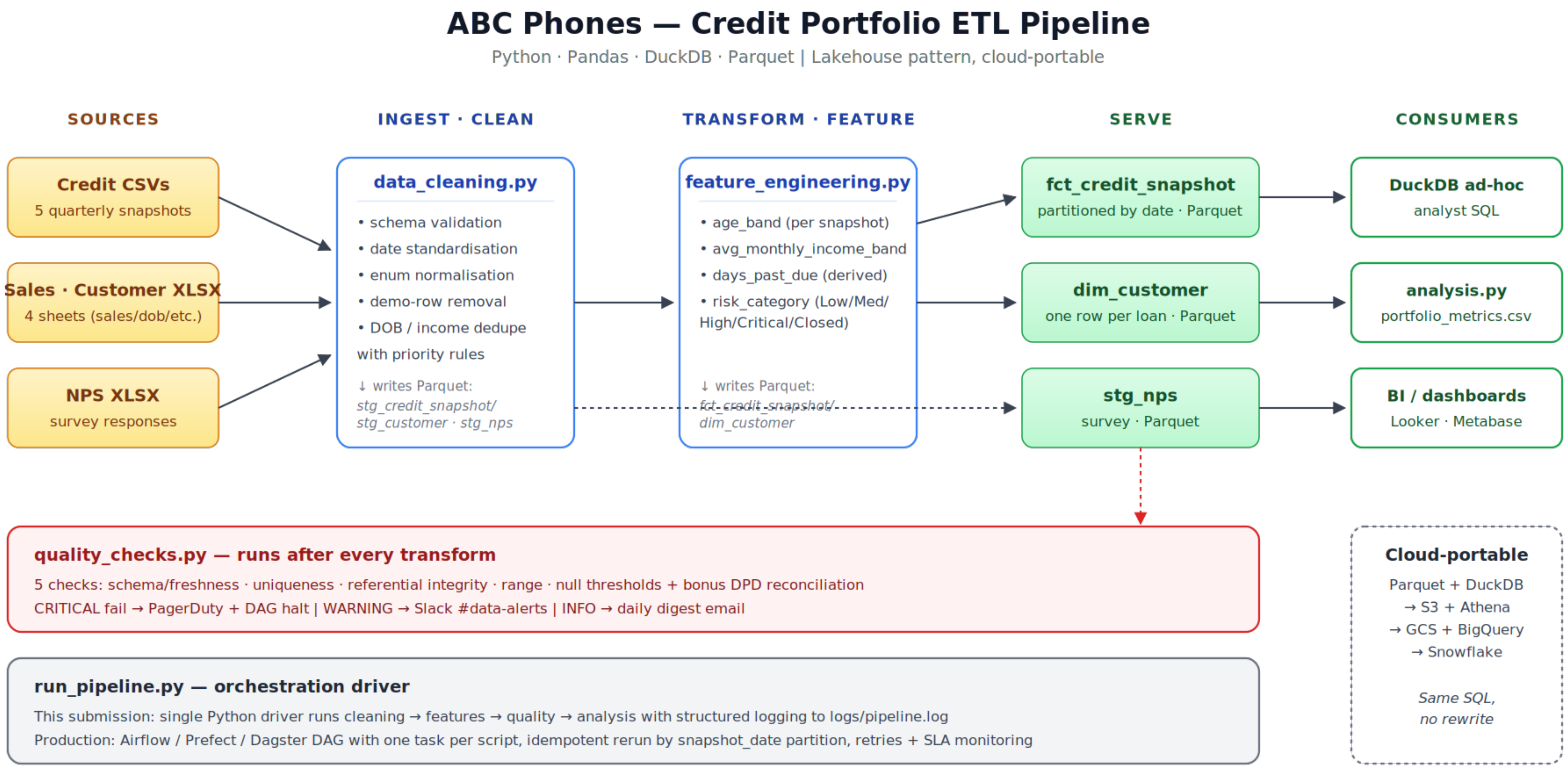
71,448 credit rows across 5 quarterly snapshots	**20,742** unique loans (1 row per loan per snapshot)	**3,985** NPS responses, 100% match credit	**~50%** credit loans w/ no DOB/income/gender

Top quality issues identified

Issue	Detail	Resolution
Filename ↔ data drift	Credit_Data_-_30-03-2025.csv contains DATE = 3/31/2025	Caught by schema_and_freshness check (CRITICAL)
No CUSTOMER_ID anywhere	Joins are loan-to-loan only via LOAN_ID / Loan Id	Documented gap; multi-loan customers not trackable
Column-name typo	DOB sheet uses 'Loan Id ' (trailing space)	Stripped on ingest
DOB conflicts	471 loans have multiple DOBs across 3 providers	Resolution rule: TransUnion → SmileID → SpinMobile, then most recent createdAt
Income column ambiguity	4 income fields likely overlap — naive sum double-counts	Use 'Received' as canonical total; documented
Excel padding	All customer sheets padded to 1,048,575 rows (Excel cap)	dropna(how='all') on ingest
Inconsistent enum casing	BALANCE_DUE_STATUS: 'Arrears' vs 'up to date' vs 'advance'	Lowercased on ingest

ETL architecture

Lakehouse pattern on a laptop today, lifts to S3+Athena / GCS+BigQuery / Snowflake tomorrow with no SQL rewrites



Key engineering decisions

Each call is small but compounds into a robust pipeline

Hive partitioning by snapshot_date

Each snapshot is its own Parquet partition. Reruns drop & replace one folder; the rest stays untouched. Idempotent without locking.

DuckDB over Pandas-only

Cleaning stays in Pandas (row-wise logic). Analytics moves to SQL via DuckDB on the Parquet files. Same SQL ports to BigQuery/Snowflake.

Pure-function stages

Every clean_* and check_* function returns (df, report). No globals, no in-place mutation. Trivially unit-testable.

Source values preserved alongside derived

We keep both days_past_due (source) and days_past_due_derived (ours). The reconciliation gap itself is a quality check — not a bug to hide.

Schema contract on ingest

Expected columns are codified in EXPECTED_CREDIT_COLS. Missing or extra cols halt the pipeline before bad data reaches analysts.

Late-arriving / reprocessing strategy

Replay any snapshot independently — `python run_pipeline.py --only cleaning` rebuilds one partition. Nothing else recomputes.

Data quality framework

Five checks + one bonus, each with severity, cadence, and a real alerting route

Check	Severity	Cadence	Alert route
1. Schema & freshness — files arrive, columns match contract, internal DATE = filename	CRITICAL	real-time	PagerDuty + halt DAG
2. Uniqueness of (loan_id, snapshot_date)	CRITICAL	daily	PagerDuty + halt DAG
3. Referential integrity — every credit/NPS loan_id exists in dim_customer	WARNING	daily	Slack #data-alerts
4. Range checks — age 18-100, income > 0, NPS 0-10, DPD ≥ 0	WARNING	daily	Slack #data-alerts
5. Null thresholds — critical fields < 1%, soft fields < 60%	WARNING	daily	Slack #data-alerts
+ DPD reconciliation — derived vs source within 7-day tolerance	INFO	weekly	Daily digest email

REAL EXAMPLE FROM THE DATA

Filename ⇔ internal date mismatch

`Credit_Data_-_30-03-2025.csv` contains an internal `DATE` column with value `3/31/2025` — silent date drift that would land in PAR-30 quarter-end metrics.

→ Caught automatically by check #1 with severity=CRITICAL on first run. Alert routed to PagerDuty; DAG halted before downstream tasks ran.

Feature engineering

Four derived fields per the brief; risk_category is where the judgement shows

Four required features

`age_band`

from DOB at each reporting date — 18-25, 26-35, 36-45, 46-55, 55+, plus explicit 'Unknown' bucket so coverage stays visible

`avg_monthly_income_band`

received_total ÷ duration_months → 8 bands per brief. 'Received' used as canonical total to avoid sub-channel double-counting

`days_past_due`

(reporting_date − next_invoice_date), clamped to 0 when no arrears. Source value retained for reconciliation

`risk_category`

the credit-risk judgement call → see right

risk_category logic

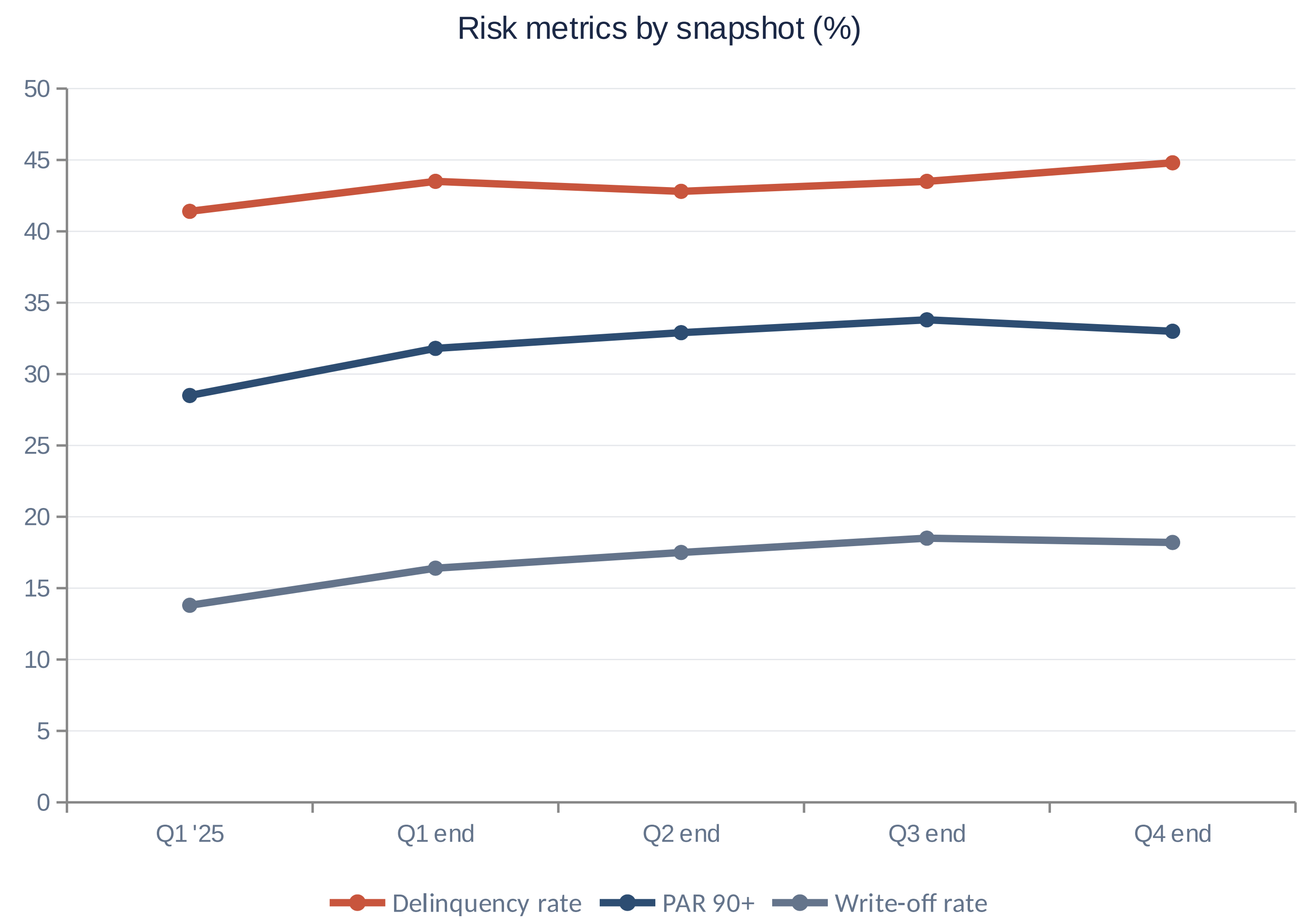
Standard PAR-bucket framework adapted to ABC Phones' status taxonomy

CRITICAL	`Write Off / Lost Write Off  OR  DPD ≥ 90`
HIGH	`DPD 31-89  OR  L2 ∈ {PAR 30, FMD}`
MEDIUM	`DPD 1-30  OR  L2 ∈ {PAR 7, FPD, Inactive}`
LOW	`DPD = 0  AND  L2 = Active`
CLOSED	`L2 = Paid Off  (out of active book)`
RETURNED	`L2 = Return  (out of active book)`

Mirrors IFRS-9 staging logic and standard microfinance PAR-bucketing.

Portfolio health: book grew fast, quality slipped

Five quarterly snapshots, 2025-01-01 through 2025-12-30



59.5%

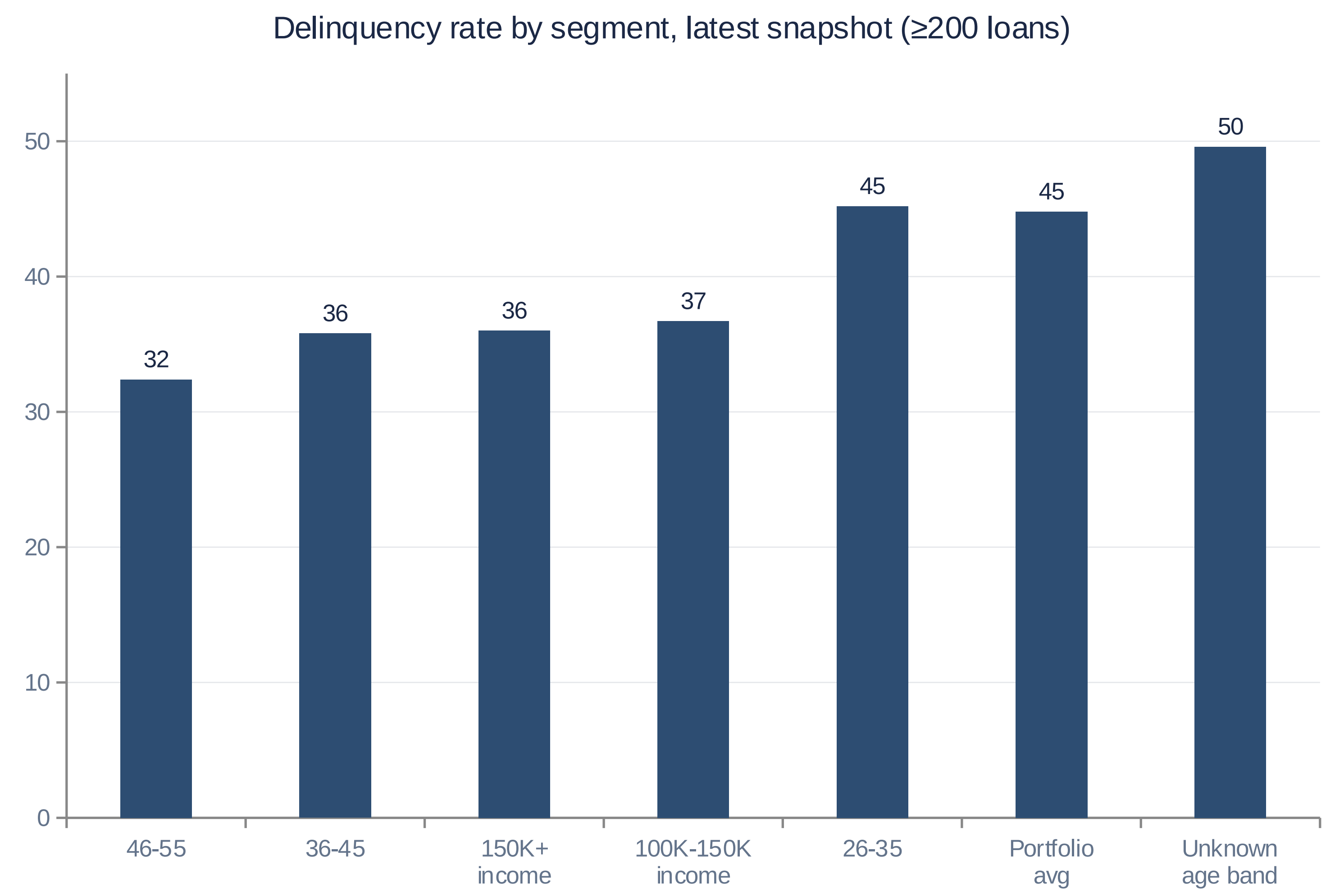
Active-loan collection rate (latest) — down from 69.4% Q1

Take-aways

- Book size more than doubled in 2025 (133% growth in active loans).
- Delinquency, PAR 90, and write-off rates all climbed in parallel — not a one-off.
- Collection rate fell 10 percentage points — the leading indicator.
- Pattern is consistent with rapid origination outpacing underwriting controls.

Where the risk really lives

Loans without complete customer profiles default ~5pp more than the rest of the book



THE FINDING

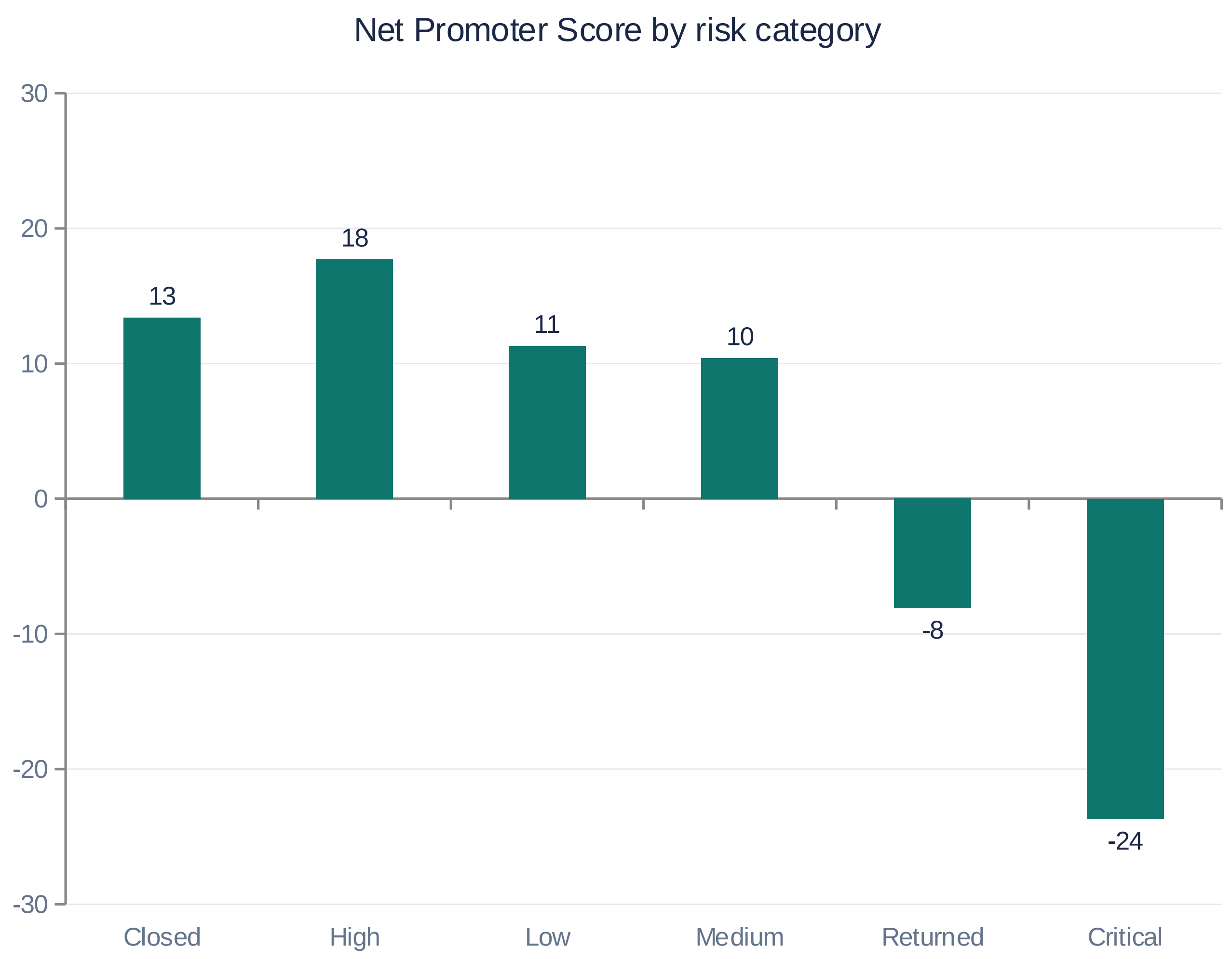
Loans with no DOB on file (46% of latest book) default at 49.6% — almost 5pp above portfolio average.

Loans WITH complete profiles (older borrowers, higher income) consistently perform 8-12pp better.

This isn't only a data gap — it's a credit-decisioning gap. The pipeline approves loans without complete profiles, and those loans default more.

Credit × NPS — and one fix that helps both

Each NPS response joined to the customer's most recent credit snapshot



RECOMMENDATION

Fix the lock-after-payment issue

12-17% of NPS respondents across every risk segment report *'phone locked despite paying on time'*

This drives **both** outcomes the business cares about: customer dissatisfaction (Critical-risk NPS = -23.7) AND avoidable arrears flags when the locking system runs ahead of payment reflection.

Fix: a 24-hour grace window before remote-locking, plus a reconciliation job that lifts locks within 1 hour of payment confirmation.

→ improves NPS and reduces 'fake' delinquency simultaneously — a rare win-win in collections design.

Where this goes next

Three data improvements ABC Phones should make + the production-roadmap delta

DATA IMPROVEMENTS

1 Introduce a stable customer_id

Hash of national ID or phone, propagated from KYC into credit and NPS. Unlocks lifetime-value, repeat-borrower risk, and household-level analysis.

2 Standardise on ISO-8601 + a YAML schema spec

Versioned alongside the data; documents the unit of every numeric column (Duration in months, balance in KES, etc.). Eliminates ambiguity discussed in Slide 2.

3 Replace daily snapshots with an event stream

Emit payment_made, status_changed, arrears_updated events; derive snapshots downstream. Removes the filename-vs-internal-date drift class of bugs entirely.

PRODUCTION ROADMAP DELTA

Orchestration	run_pipeline.py → Airflow / Prefect / Dagster DAG; one task per stage; SLAs
Storage	Local Parquet → S3 / GCS, same Hive partitioning
Compute	DuckDB → Athena or BigQuery — same SQL, no rewrite
Transforms	Python + SQL → dbt (lineage, docs, tests for free)
Quality	Custom Python checks → Great Expectations / dbt tests / Soda
Alerting	Mocked log routes → PagerDuty + Slack SDKs, on-call rota
Lineage	Add → OpenLineage events from each task